Ramidi Chinmayee

co23btech11019@iith.ac.in | ramidichinmayee@gmail.com | +91 8309562809 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.56
Sri Chaitanya Junior College	Class XII (Telangana State Board)	2023	98.60%
Sri Chaitanya High School	Class X (Board of Secondary Education, Telangana)	2021	10.00%

WORK EXPERIENCE

Technology & AI Engineering Intern - Morgan Stanley

AI Agents, Model Context Protocol (MCP), Kafka, LLMs, Infrastructure Automation

- Developed **autonomous AI-agent workflows using Atlassian MCP** to automate Kafka topic access control and **PRO-ID whitelisting across 4 infrastructure environments**, fully automating ~30 requests daily.
- Integrated **LLM-driven tools** to parse, structure, and validate **JIRA ticket data**, enabling reliable automated access-provisioning decisions and reducing manual processing effort.
- Engineered **Git/GitHub automation pipelines** to apply repository configurations, commit changes, and raise verified Pull Requests, **reducing request processing time by 95% from 20 minutes to 1 minute.**

PROJECTS

High-Performance HTTP Load Balancer

Personal Project | C++, HTTP, Networking, Multithreading, Load Balancing

- Built a **concurrent HTTP load balancer** to distribute client requests across multiple backend servers.
- Implemented **round-robin routing and backend health checks** for failure-aware request distribution.
- Benchmarked **throughput and latency** under varying client loads to analyze scalability and system performance.

Multithreading, Semaphores & Locks

Course Project - Operating Systems | C++, Threading, Synchronization, Mutual Exclusion Algorithms

- Built multithreaded Sudoku validators using **static allocation (Chunk and Mixed)**, benchmarking performance.
- Implemented Sudoku validation with **TAS, CAS, and Bounded CAS**, analyzing synchronization overhead.
- Solved the **Producer-Consumer problem** using semaphores and locks, benchmarking synchronization performance.

Codebase Intelligence & Debugging Assistant

Personal Project | Python, RAG, LLMs, Embeddings, Vector Database

- Built a **codebase ingestion pipeline** to parse, chunk, embed, and index source files for semantic retrieval.
- Implemented a **RAG pipeline with LLM-based tool calling** to retrieve relevant code and answer debugging queries.
- Designed **agentic workflows** for code search, file inspection, and dependency analysis across repositories.

FOML Mini Machine Learning Library

Course Project - Foundations of Machine Learning | Python, Autograd, Neural Networks, SGD

- Built an **ML library from scratch** with preprocessing, logistic regression, and an autograd-based neural network.
- Implemented **layers, activations, losses, and SGD**, enabling end-to-end training without external ML frameworks.
- Applied and tuned **ML architectures on Fashion-MNIST**, comparing generalization using validation and test accuracy.

SKILLS

Programming: Python, C++, C, SQL, Julia, MATLAB

Machine Learning: TensorFlow, Keras, PyTorch, scikit-learn, XGBoost, NumPy, Pandas

AI / GenAI: OpenCV, Face Recognition, RetinaFace, ArcFace, FAISS, Sentence-Transformers, RAG, LLMs, LangChain

Systems & Development: Multithreading, Synchronization, Concurrency, HTTP, Networking, Kafka, MCP, Git, GitHub, MySQL

Tools & Visualization: Jupyter, Streamlit, SciPy, LaTeX, Matplotlib

RELEVANT COURSEWORK

Computer Science: Data Structures & Algorithms, Operating Systems, Foundations of Machine Learning, Intro to Programming

Mathematics: Probability & Stochastic Processes, Linear Algebra, Numerical Methods, Statistics, Optimization Techniques

Computational Engineering: Fundamentals of Scientific Computing, Computer-Aided Numerical Methods I & II, Computational Methods in Materials Science, Electronic Structure & Atomistic Modeling

AI & Data: Machine Learning in Materials Science, Big Data Biology & Biological Databases, AI for Drug Discovery, Biostatistics

SCHOLASTIC ACHIEVEMENTS

All India Rank of **4843** in JEE Advanced

All India Rank of **6285** in JEE Mains

Ranked in the **Top 150** out of 1 lakh students nationwide and qualified for **RMO**

Awarded **NTSE Scholarship** by the **Government of India** for clearing the National Talent Search Examination

POSITIONS OF RESPONSIBILITY

Head of Finance, Elan & nVision, annual techno-cultural fest of IIT-Hyderabad

- Leading the Finance domain to manage funds of over 50 Lakh INR

- Managing all financial activities including budgeting, reimbursements, and coordination with other teams

Coordinator, Elan & nVision, Sponsorship Domain

EXTRA-CURRICULAR

- Graphite Art, Portrait Sketching, Badminton
- NSS Volunteer